Section 5 Case Study

Qi He

2026-02-21

1. Data Processing

```r
# ==========================================================
# Case 1 Pipeline in R (Upgraded carbon story)
# Virginia / PJM / DOM
# ==========================================================

# ------------------------------
# 0. Packages
# ------------------------------
required_pkgs <- c(
  "jsonlite", "readr", "dplyr", "tidyr", "ggplot2", "scales", "stringr"
)

installed <- rownames(installed.packages())
for (p in required_pkgs) {
  if (!(p %in% installed)) install.packages(p)
}

library(jsonlite)
library(readr)
```

```
## Warning: package 'readr' was built under R version 4.5.2
```

```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(tidyr)
library(ggplot2)
library(scales)
```

```
##
## Attaching package: 'scales'

## The following object is masked from 'package:readr':
##
##     col_factor
```

```r
library(stringr)

# ----------------------------
# 1. Set working directory
# ----------------------------
setwd("D:/Google Drive/R/case1_pipeline_package/case1_pipeline")

getwd()
```

```
## [1] "D:/Google Drive/R/case1_pipeline_package/case1_pipeline"
```

```r
list.files()
```

```
##  [1] "build_case1_hourly_dataset.py"
##  [2] "case1_hourly_dataset_built_from_R.csv"
##  [3] "case1_option_hourly_components_from_R.csv"
##  [4] "case1_option_hourly_components_upgraded.csv"
##  [5] "case1_option_summary_from_R.csv"
##  [6] "case1_option_summary_upgraded.csv"
##  [7] "case1_parameters.json"
##  [8] "compute_case1_crcc.py"
##  [9] "desktop.ini"
## [10] "Figure_5_1_CRCC_stacked_bar.pdf"
## [11] "Figure_5_1_CRCC_stacked_bar.png"
## [12] "Figure_5_1_CRCC_stacked_bar_labeled.png"
## [13] "Figure_5_1_CRCC_stacked_bar_upgraded.pdf"
## [14] "Figure_5_1_CRCC_stacked_bar_upgraded.png"
## [15] "Figure_5_2_BSM_vs_CRCC.pdf"
## [16] "Figure_5_2_BSM_vs_CRCC.png"
## [17] "figure5_2_bsm_results.csv"
## [18] "figure5_2_comparison_table.csv"
## [19] "figure5_2_crcc_results.csv"
## [20] "figure5_2_design_gap.csv"
## [21] "raw_template_hourly_emissions.csv"
## [22] "raw_template_pjm_lmp_dom.csv"
## [23] "raw_template_pjm_stress_hours.csv"
## [24] "README_case1_pipeline.md"
## [25] "sample_case1_hourly_2025.csv"
## [26] "sample_case1_option_hourly_components.csv"
## [27] "sample_case1_option_summary.csv"
```

```
## [28] "section-5-case-study.html"
## [29] "section-5-case-study.Rmd"
## [30] "section-5-case-study.tex"
## [31] "section 5 case study.Rmd"
```

```r
# ------------------------------
# 2. Read parameters and hourly data
# ------------------------------
params <- fromJSON("case1_parameters.json", simplifyVector = FALSE)

hourly <- read_csv("sample_case1_hourly_2025.csv", show_col_types = FALSE) %>%
  mutate(timestamp = as.POSIXct(timestamp, format = "%Y-%m-%d %H:%M:%S", tz = "UTC"))

# ------------------------------
# 3. Basic checks
# ------------------------------
required_cols <- c(
  "timestamp",
  "lmp_usd_per_mwh",
  "stress_indicator",
  "emissions_tco2_per_mwh",
  "it_load_mw",
  "pue",
  "eta_compute_per_mwh_it",
  "carbon_shadow_price_usd_per_tco2",
  "reliability_valuation_usd_per_compute_unit",
  "location_based_emissions_baseline_tco2_per_mwh"
)

missing_cols <- setdiff(required_cols, names(hourly))
if (length(missing_cols) > 0) {
  stop(paste("Missing required columns:", paste(missing_cols, collapse = ", ")))
}

# ------------------------------
# 4. Extract planning parameters
# ------------------------------
pp <- params$planning_parameters

it_load_mw <- as.numeric(pp$it_load_mw)
pue <- as.numeric(pp$pue)
eta <- as.numeric(pp$eta_compute_per_mwh_it)
lambda <- as.numeric(pp$carbon_shadow_price_usd_per_tco2)
nu <- as.numeric(pp$reliability_valuation_usd_per_compute_unit)
g_lb <- as.numeric(pp$location_based_emissions_baseline_tco2_per_mwh)
```

2. Comparative Experiment (structured scenario experiment)

```r
# ------------------------------
# 5. Define upgraded option assumptions
# ------------------------------
# We keep the same two stylized designs, but now explicitly allow
```

```r
# option-specific carbon exposure through carbon_exposure_multiplier.
#
# Interpretation:
# - Connect-and-Manage: full exposure to hourly marginal emissions
# - Associated supply / reduced exposure: lower effective carbon exposure
#
# can tune these values later if needed.

option_assumptions <- tibble::tibble(
  option = c("Connect-and-Manage", "Associated_supply_reduced_exposure"),
  option_label = c("Connect-and-Manage", "Associated supply / reduced exposure"),
  price_multiplier = c(1.00, 1.08),
  carbon_exposure_multiplier = c(1.00, 0.70),
  stress_severity_kappa = c(0.15, 0.03)
)

print(option_assumptions)
```

```
## # A tibble: 2 x 5
##   option                    option_label price_multiplier carbon_exposure_mult~1
##   <chr>                     <chr>                  <dbl>                  <dbl>
## 1 Connect-and-Manage        Connect-and~            1                       1
## 2 Associated_supply_reduce~ Associated ~            1.08                    0.7
## # i abbreviated name: 1: carbon_exposure_multiplier
## # i 1 more variable: stress_severity_kappa <dbl>
```

```r
# ----------------------------
# 6. Computation function
# ----------------------------
compute_option_components <- function(df, option_name, option_label,
                                      price_multiplier,
                                      carbon_exposure_multiplier,
                                      stress_severity_kappa,
                                      it_load_mw, pue, eta, lambda, nu) {

  df_out <- df %>%
    mutate(
      option = option_name,
      option_label = option_label,
      price_multiplier = price_multiplier,
      carbon_exposure_multiplier = carbon_exposure_multiplier,
      stress_severity_kappa = stress_severity_kappa,

      # Core physical/service mapping
      facility_electricity_mwh = it_load_mw * pue,
      potential_compute = it_load_mw * eta,
      availability_alpha = 1 - stress_severity_kappa * stress_indicator,
      delivered_compute = availability_alpha * potential_compute,

      # Effective decision-facing emissions intensity
      effective_emissions_tco2_per_mwh = carbon_exposure_multiplier * emissions_tco2_per_mwh,

      # Component computations
```

```r
      electricity_component_usd =
        lmp_usd_per_mwh * price_multiplier * facility_electricity_mwh,

      carbon_component_usd =
        lambda * effective_emissions_tco2_per_mwh * facility_electricity_mwh,

      reliability_component_usd =
        nu * (1 - availability_alpha) * potential_compute,

      crcc_hourly_numerator_usd =
        electricity_component_usd +
        carbon_component_usd +
        reliability_component_usd
    )

  return(df_out)
}

# ------------------------------
# 7. Compute hourly outputs for all options
# ------------------------------
hourly_components_list <- lapply(seq_len(nrow(option_assumptions)), function(i) {
  compute_option_components(
    df = hourly,
    option_name = option_assumptions$option[i],
    option_label = option_assumptions$option_label[i],
    price_multiplier = option_assumptions$price_multiplier[i],
    carbon_exposure_multiplier = option_assumptions$carbon_exposure_multiplier[i],
    stress_severity_kappa = option_assumptions$stress_severity_kappa[i],
    it_load_mw = it_load_mw,
    pue = pue,
    eta = eta,
    lambda = lambda,
    nu = nu
  )
})

hourly_components <- bind_rows(hourly_components_list)

# ------------------------------
# 8. Create summary table
# ------------------------------
option_summary <- hourly_components %>%
  group_by(option, option_label) %>%
  summarise(
    electricity_component_usd_per_delivered_compute =
      sum(electricity_component_usd) / sum(delivered_compute),

    carbon_component_usd_per_delivered_compute =
      sum(carbon_component_usd) / sum(delivered_compute),

    reliability_component_usd_per_delivered_compute =
      sum(reliability_component_usd) / sum(delivered_compute),
```

```r
    crcc_total_usd_per_delivered_compute =
      sum(crcc_hourly_numerator_usd) / sum(delivered_compute),

    expected_delivered_compute_total = sum(delivered_compute),
    total_potential_compute = sum(potential_compute),
    stress_hours = sum(stress_indicator),
    avg_price_usd_per_mwh = mean(lmp_usd_per_mwh),
    avg_base_emissions_tco2_per_mwh = mean(emissions_tco2_per_mwh),
    avg_effective_emissions_tco2_per_mwh = mean(effective_emissions_tco2_per_mwh),
    .groups = "drop"
  ) %>%
  mutate(across(where(is.numeric), ~ round(.x, 4)))

print(option_summary)
```

```
## # A tibble: 2 x 12
##   option              option_label electricity_componen~1 carbon_component_usd~2
##   <chr>               <chr>                          <dbl>                  <dbl>
## 1 Associated_supply_~ Associated ~                    65.7                   17.5
## 2 Connect-and-Manage  Connect-and~                    60.9                   25.0
## # i abbreviated names: 1: electricity_component_usd_per_delivered_compute,
## #   2: carbon_component_usd_per_delivered_compute
## # i 8 more variables: reliability_component_usd_per_delivered_compute <dbl>,
## #   crcc_total_usd_per_delivered_compute <dbl>,
## #   expected_delivered_compute_total <dbl>, total_potential_compute <dbl>,
## #   stress_hours <dbl>, avg_price_usd_per_mwh <dbl>,
## #   avg_base_emissions_tco2_per_mwh <dbl>, ...
```

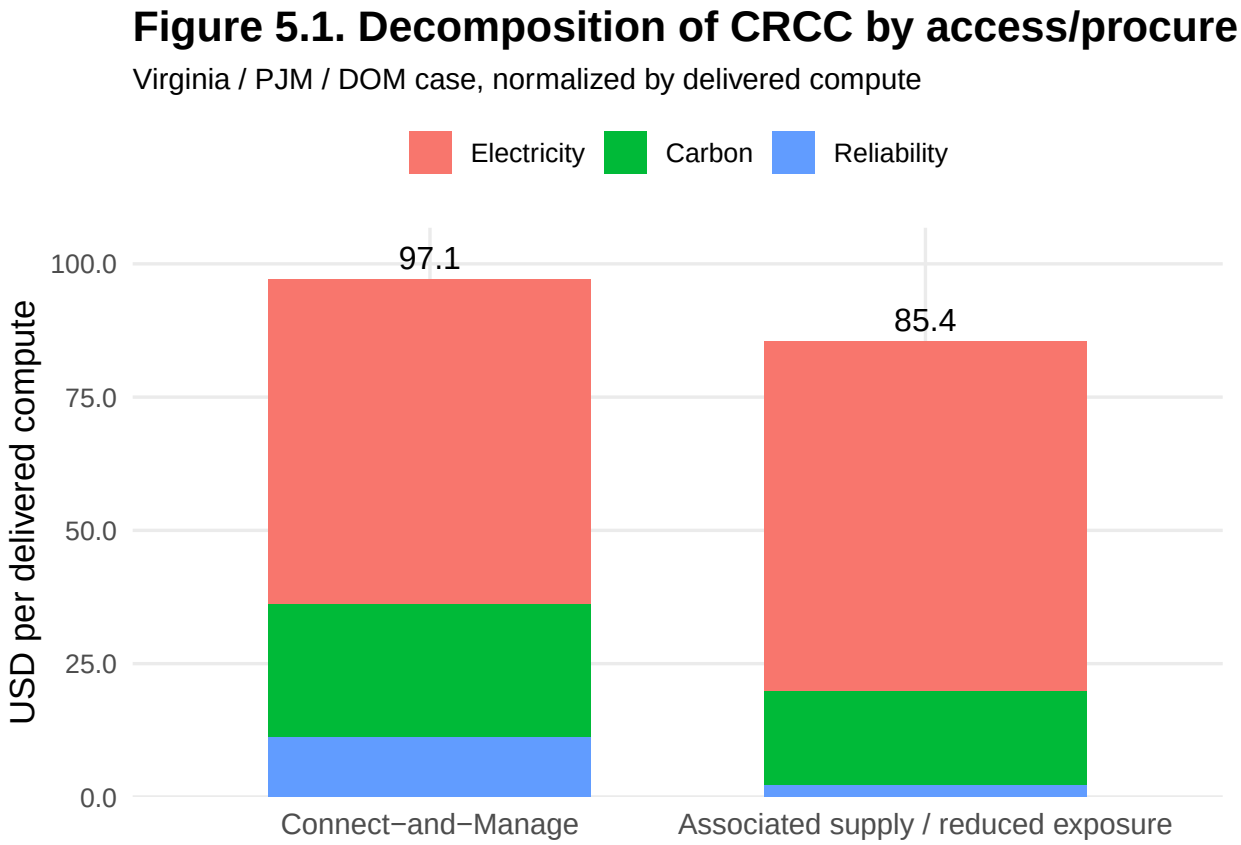
```r
# ----------------------------
# 9. Export output data
# ----------------------------
write_csv(hourly_components, "case1_option_hourly_components_upgraded.csv")
write_csv(option_summary, "case1_option_summary_upgraded.csv")
```

2.1. Figure 5.1

```r
# 10. Prepare plotting dataset for Figure 5.1
# ----------------------------
plot_data <- option_summary %>%
  select(
    option_label,
    electricity_component_usd_per_delivered_compute,
    carbon_component_usd_per_delivered_compute,
    reliability_component_usd_per_delivered_compute,
    crcc_total_usd_per_delivered_compute
  ) %>%
  pivot_longer(
    cols = c(
      electricity_component_usd_per_delivered_compute,
      carbon_component_usd_per_delivered_compute,
```

```r
      reliability_component_usd_per_delivered_compute
    ),
    names_to = "component",
    values_to = "value"
  ) %>%
  mutate(
    component = recode(
      component,
      "electricity_component_usd_per_delivered_compute" = "Electricity",
      "carbon_component_usd_per_delivered_compute" = "Carbon",
      "reliability_component_usd_per_delivered_compute" = "Reliability"
    )
  )

plot_data$component <- factor(
  plot_data$component,
  levels = c("Electricity", "Carbon", "Reliability")
)

plot_data$option_label <- factor(
  plot_data$option_label,
  levels = c("Connect-and-Manage", "Associated supply / reduced exposure")
)

plot_totals <- option_summary %>%
  mutate(
    option_label = factor(
      option_label,
      levels = c("Connect-and-Manage", "Associated supply / reduced exposure")
    )
  )

# ------------------------------
# 11. Draw Figure 5.1 with total labels
# ------------------------------
fig_5_1 <- ggplot(plot_data, aes(x = option_label, y = value, fill = component)) +
  geom_col(width = 0.65) +
  geom_text(
    data = plot_totals,
    aes(
      x = option_label,
      y = crcc_total_usd_per_delivered_compute,
      label = round(crcc_total_usd_per_delivered_compute, 1)
    ),
    inherit.aes = FALSE,
    vjust = -0.45,
    size = 4.2
  ) +
  scale_y_continuous(
    labels = number_format(accuracy = 0.1),
    expand = expansion(mult = c(0, 0.10))
  ) +
  labs(
```

```r
    title = "Figure 5.1. Decomposition of CRCC by access/procurement design",
    subtitle = "Virginia / PJM / DOM case, normalized by delivered compute",
    x = NULL,
    y = "USD per delivered compute",
    fill = NULL
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold"),
    plot.subtitle = element_text(size = 11),
    axis.text.x = element_text(size = 11),
    legend.position = "top",
    panel.grid.minor = element_blank()
  )

print(fig_5_1)
```




```r
# ----------------------------
# 12. Save figure
# ----------------------------
ggsave(
  filename = "Figure_5_1_CRCC_stacked_bar_upgraded.png",
  plot = fig_5_1,
  width = 8.5,
  height = 5.5,
```

```r
  dpi = 300
)

ggsave(
  filename = "Figure_5_1_CRCC_stacked_bar_upgraded.pdf",
  plot = fig_5_1,
  width = 8.5,
  height = 5.5
)
```

2.2. Additional Steps

```r
# 13. Optional diagnostics
# ----------------------------
# Check total stress hours
hourly %>%
  summarise(total_stress_hours = sum(stress_indicator))
```

```
## # A tibble: 1 x 1
##   total_stress_hours
##                <dbl>
## 1                 65
```

```r
# Check yearly benchmark averages
hourly %>%
  summarise(
    avg_price = mean(lmp_usd_per_mwh),
    avg_emissions = mean(emissions_tco2_per_mwh),
    lb_emissions_baseline = mean(location_based_emissions_baseline_tco2_per_mwh)
  )
```

```
## # A tibble: 1 x 3
##   avg_price avg_emissions lb_emissions_baseline
##       <dbl>         <dbl>                 <dbl>
## 1      50.7         0.260                 0.268
```

3. Baseline Screening Metric (BSM) vs CRCC

```r
# ==========================================================
# Figure 5.2 in R
# Baseline Screening Metric (BSM) vs CRCC
# Virginia / PJM / DOM Case
# ==========================================================

# ----------------------------
# 0. Packages
# ----------------------------
required_pkgs <- c(
  "jsonlite", "readr", "dplyr", "tidyr", "ggplot2", "scales", "stringr"
```

```r
)

installed <- rownames(installed.packages())
for (p in required_pkgs) {
  if (!(p %in% installed)) install.packages(p)
}

library(jsonlite)
library(readr)
library(dplyr)
library(tidyr)
library(ggplot2)
library(scales)
library(stringr)

# ------------------------------
# 1. Set working directory
# ------------------------------
setwd("D:/Google Drive/R/case1_pipeline_package/case1_pipeline")

getwd()
```

```
## [1] "D:/Google Drive/R/case1_pipeline_package/case1_pipeline"
```

```r
list.files()
```

```
##  [1] "build_case1_hourly_dataset.py"
##  [2] "case1_hourly_dataset_built_from_R.csv"
##  [3] "case1_option_hourly_components_from_R.csv"
##  [4] "case1_option_hourly_components_upgraded.csv"
##  [5] "case1_option_summary_from_R.csv"
##  [6] "case1_option_summary_upgraded.csv"
##  [7] "case1_parameters.json"
##  [8] "compute_case1_crcc.py"
##  [9] "desktop.ini"
## [10] "Figure_5_1_CRCC_stacked_bar.pdf"
## [11] "Figure_5_1_CRCC_stacked_bar.png"
## [12] "Figure_5_1_CRCC_stacked_bar_labeled.png"
## [13] "Figure_5_1_CRCC_stacked_bar_upgraded.pdf"
## [14] "Figure_5_1_CRCC_stacked_bar_upgraded.png"
## [15] "Figure_5_2_BSM_vs_CRCC.pdf"
## [16] "Figure_5_2_BSM_vs_CRCC.png"
## [17] "figure5_2_bsm_results.csv"
## [18] "figure5_2_comparison_table.csv"
## [19] "figure5_2_crcc_results.csv"
## [20] "figure5_2_design_gap.csv"
## [21] "raw_template_hourly_emissions.csv"
## [22] "raw_template_pjm_lmp_dom.csv"
## [23] "raw_template_pjm_stress_hours.csv"
## [24] "README_case1_pipeline.md"
## [25] "sample_case1_hourly_2025.csv"
## [26] "sample_case1_option_hourly_components.csv"
## [27] "sample_case1_option_summary.csv"
```

```
## [28] "section-5-case-study.html"
## [29] "section-5-case-study.Rmd"
## [30] "section-5-case-study.tex"
## [31] "section-5-case-study_files"
## [32] "section 5 case study.Rmd"
```

```r
# ------------------------------
# 2. Read parameters and hourly data
# ------------------------------
params <- fromJSON("case1_parameters.json", simplifyVector = FALSE)

hourly <- read_csv("sample_case1_hourly_2025.csv", show_col_types = FALSE) %>%
  mutate(timestamp = as.POSIXct(timestamp, format = "%Y-%m-%d %H:%M:%S", tz = "UTC"))

# ------------------------------
# 3. Basic checks
# ------------------------------
required_cols <- c(
  "timestamp",
  "lmp_usd_per_mwh",
  "stress_indicator",
  "emissions_tco2_per_mwh",
  "it_load_mw",
  "pue",
  "eta_compute_per_mwh_it",
  "carbon_shadow_price_usd_per_tco2",
  "reliability_valuation_usd_per_compute_unit",
  "location_based_emissions_baseline_tco2_per_mwh"
)

missing_cols <- setdiff(required_cols, names(hourly))
if (length(missing_cols) > 0) {
  stop(paste("Missing required columns:", paste(missing_cols, collapse = ", ")))
}

# ------------------------------
# 4. Extract planning parameters
# ------------------------------
pp <- params$planning_parameters

it_load_mw <- as.numeric(pp$it_load_mw)
pue <- as.numeric(pp$pue)
eta <- as.numeric(pp$eta_compute_per_mwh_it)
lambda <- as.numeric(pp$carbon_shadow_price_usd_per_tco2)
nu <- as.numeric(pp$reliability_valuation_usd_per_compute_unit)
g_lb <- as.numeric(pp$location_based_emissions_baseline_tco2_per_mwh)

# ------------------------------
# 5. Define option assumptions
# Same logic as upgraded Figure 5.1
# ------------------------------
option_assumptions <- tibble::tibble(
  option = c("Connect-and-Manage", "Associated_supply_reduced_exposure"),
  option_label = c("Connect-and-Manage", "Associated supply / reduced exposure"),
```

```r
  price_multiplier = c(1.00, 1.08),
  carbon_exposure_multiplier = c(1.00, 0.70),
  stress_severity_kappa = c(0.15, 0.03)
)

print(option_assumptions)
```

```
## # A tibble: 2 x 5
##   option                    option_label price_multiplier carbon_exposure_mult~1
##   <chr>                     <chr>                   <dbl>                  <dbl>
## 1 Connect-and-Manage        Connect-and~             1                       1
## 2 Associated_supply_reduce~ Associated ~             1.08                    0.7
## # i abbreviated name: 1: carbon_exposure_multiplier
## # i 1 more variable: stress_severity_kappa <dbl>
```

```r
# ----------------------------
# 6. Common quantities
# ----------------------------
T_hours <- nrow(hourly)
avg_price <- mean(hourly$lmp_usd_per_mwh, na.rm = TRUE)

facility_electricity_mwh_per_hour <- it_load_mw * pue
potential_compute_per_hour <- it_load_mw * eta

sum_E <- T_hours * facility_electricity_mwh_per_hour
sum_Q_pot <- T_hours * potential_compute_per_hour

# ----------------------------
# 7. Function to compute CRCC
# ----------------------------
compute_crcc <- function(df,
                         option_name,
                         option_label,
                         price_multiplier,
                         carbon_exposure_multiplier,
                         stress_severity_kappa,
                         it_load_mw, pue, eta, lambda, nu) {

  tmp <- df %>%
    mutate(
      option = option_name,
      option_label = option_label,
      price_multiplier = price_multiplier,
      carbon_exposure_multiplier = carbon_exposure_multiplier,
      stress_severity_kappa = stress_severity_kappa,

      facility_electricity_mwh = it_load_mw * pue,
      potential_compute = it_load_mw * eta,
      availability_alpha = 1 - stress_severity_kappa * stress_indicator,
      delivered_compute = availability_alpha * potential_compute,

      effective_emissions_tco2_per_mwh = carbon_exposure_multiplier * emissions_tco2_per_mwh,
```

```r
      electricity_component_usd =
        lmp_usd_per_mwh * price_multiplier * facility_electricity_mwh,

      carbon_component_usd =
        lambda * effective_emissions_tco2_per_mwh * facility_electricity_mwh,

      reliability_component_usd =
        nu * (1 - availability_alpha) * potential_compute,

      crcc_hourly_numerator_usd =
        electricity_component_usd +
        carbon_component_usd +
        reliability_component_usd
    )

  out <- tmp %>%
    summarise(
      option = first(option),
      option_label = first(option_label),
      CRCC =
        sum(crcc_hourly_numerator_usd) / sum(delivered_compute),
      electricity_component_crcc =
        sum(electricity_component_usd) / sum(delivered_compute),
      carbon_component_crcc =
        sum(carbon_component_usd) / sum(delivered_compute),
      reliability_component_crcc =
        sum(reliability_component_usd) / sum(delivered_compute),
      delivered_compute_total = sum(delivered_compute),
      potential_compute_total = sum(potential_compute),
      stress_hours = sum(stress_indicator),
      avg_effective_emissions_tco2_per_mwh = mean(effective_emissions_tco2_per_mwh)
    )

  return(out)
}

# ------------------------------
# 8. Compute CRCC results for all options
# ------------------------------
crcc_results <- bind_rows(lapply(seq_len(nrow(option_assumptions)), function(i) {
  compute_crcc(
    df = hourly,
    option_name = option_assumptions$option[i],
    option_label = option_assumptions$option_label[i],
    price_multiplier = option_assumptions$price_multiplier[i],
    carbon_exposure_multiplier = option_assumptions$carbon_exposure_multiplier[i],
    stress_severity_kappa = option_assumptions$stress_severity_kappa[i],
    it_load_mw = it_load_mw,
    pue = pue,
    eta = eta,
    lambda = lambda,
    nu = nu
  )
```

```r
})) %>%
  mutate(across(where(is.numeric), ~ round(.x, 4)))

print(crcc_results)
```

```
## # A tibble: 2 x 10
##   option         option_label  CRCC electricity_componen~1 carbon_component_crcc
##   <chr>          <chr>        <dbl>                  <dbl>                 <dbl>
## 1 Connect-and-M~ Connect-and~  97.1                   60.9                  25.0
## 2 Associated_su~ Associated ~  85.4                   65.7                  17.5
## # i abbreviated name: 1: electricity_component_crcc
## # i 5 more variables: reliability_component_crcc <dbl>,
## #   delivered_compute_total <dbl>, potential_compute_total <dbl>,
## #   stress_hours <dbl>, avg_effective_emissions_tco2_per_mwh <dbl>
```

```r
# ----------------------------
# 9. Compute BSM (Baseline Screening Metric)
# Formula:
# BSM(a,w) = [avg_price * m_(a,w) * sum E + lambda * g_LB * sum E] / sum Q_pot
#
# It intentionally ignores:
# - hourly price variation in the evaluator
# - option-specific decision-facing carbon exposure
# - stress-hour correction in the denominator
# - explicit reliability penalty
# ----------------------------
bsm_results <- option_assumptions %>%
  transmute(
    option,
    option_label,
    BSM =
      (
        avg_price * price_multiplier * sum_E +
          lambda * g_lb * sum_E
      ) / sum_Q_pot,
    avg_price_usd_per_mwh = avg_price,
    location_based_emissions_tco2_per_mwh = g_lb
  ) %>%
  mutate(across(where(is.numeric), ~ round(.x, 4)))

print(bsm_results)
```

```
## # A tibble: 2 x 5
##   option         option_label   BSM avg_price_usd_per_mwh location_based_emiss~1
##   <chr>          <chr>        <dbl>                 <dbl>                  <dbl>
## 1 Connect-and-M~ Connect-and~  86.5                  50.7                  0.268
## 2 Associated_su~ Associated ~  91.4                  50.7                  0.268
## # i abbreviated name: 1: location_based_emissions_tco2_per_mwh
```

```r
# ----------------------------
# 10. Merge evaluator results
# ----------------------------
```

```r
comparison_table <- crcc_results %>%
  select(option, option_label, CRCC,
         electricity_component_crcc,
         carbon_component_crcc,
         reliability_component_crcc,
         delivered_compute_total,
         potential_compute_total,
         stress_hours) %>%
  left_join(bsm_results, by = c("option", "option_label")) %>%
  mutate(
    BSM_minus_CRCC = round(BSM - CRCC, 4)
  )

print(comparison_table)
```

```
## # A tibble: 2 x 13
##   option         option_label  CRCC electricity_componen~1 carbon_component_crcc
##   <chr>          <chr>        <dbl>                  <dbl>                 <dbl>
## 1 Connect-and-M~ Connect-and~  97.1                   60.9                  25.0
## 2 Associated_su~ Associated ~  85.4                   65.7                  17.5
## # i abbreviated name: 1: electricity_component_crcc
## # i 8 more variables: reliability_component_crcc <dbl>,
## #   delivered_compute_total <dbl>, potential_compute_total <dbl>,
## #   stress_hours <dbl>, BSM <dbl>, avg_price_usd_per_mwh <dbl>,
## #   location_based_emissions_tco2_per_mwh <dbl>, BSM_minus_CRCC <dbl>
```

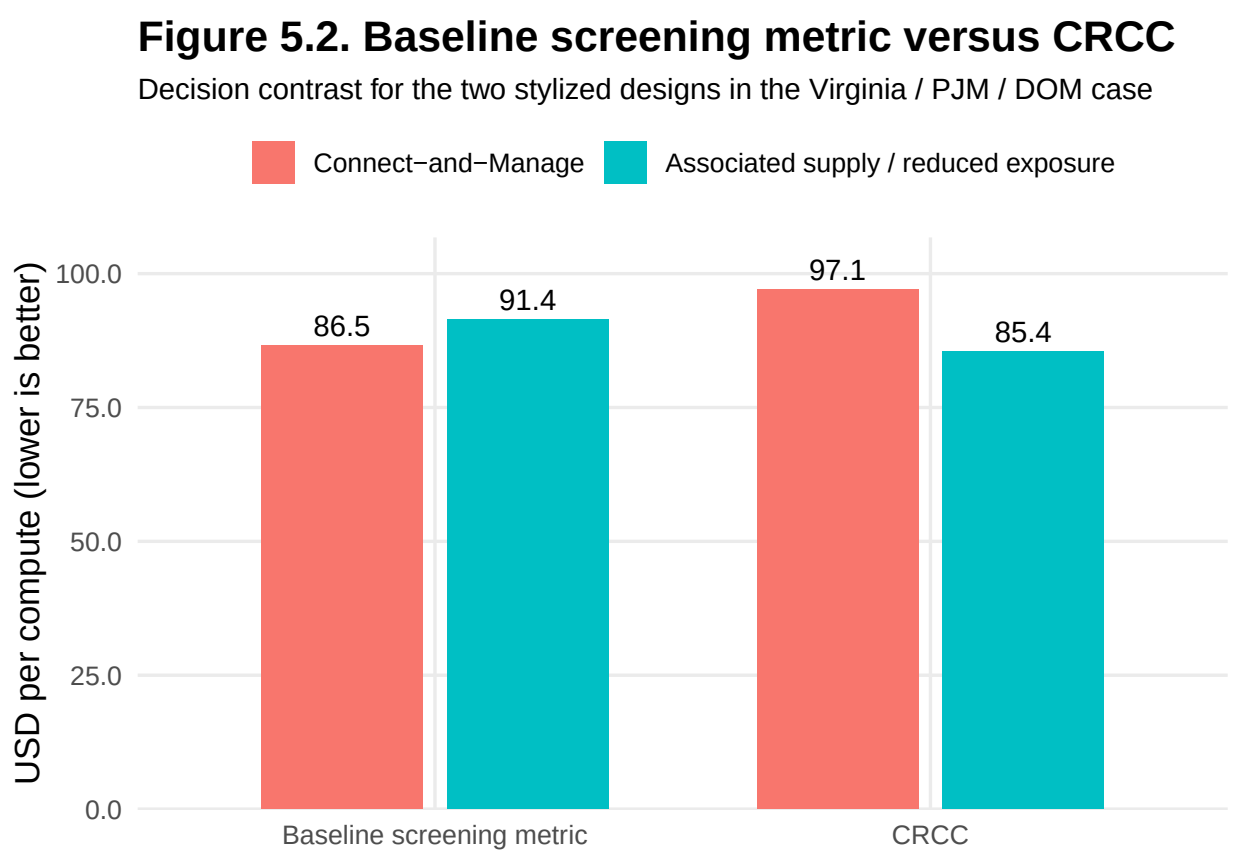
```r
# ----------------------------
# 11. Design-gap calculations
# Reduced exposure minus Connect-and-Manage
# ----------------------------
cm_bsm <- comparison_table %>%
  filter(option == "Connect-and-Manage") %>%
  pull(BSM)

re_bsm <- comparison_table %>%
  filter(option == "Associated_supply_reduced_exposure") %>%
  pull(BSM)

cm_crcc <- comparison_table %>%
  filter(option == "Connect-and-Manage") %>%
  pull(CRCC)

re_crcc <- comparison_table %>%
  filter(option == "Associated_supply_reduced_exposure") %>%
  pull(CRCC)

design_gap <- tibble(
  delta_BSM = round(re_bsm - cm_bsm, 4),
  delta_CRCC = round(re_crcc - cm_crcc, 4),
  ranking_under_BSM = ifelse(re_bsm < cm_bsm,
                             "Reduced exposure preferred",
                             "Connect-and-Manage preferred"),
  ranking_under_CRCC = ifelse(re_crcc < cm_crcc,
```

```r
                              "Reduced exposure preferred",
                              "Connect-and-Manage preferred")
)

print(design_gap)
```

```
## # A tibble: 1 x 4
##   delta_BSM delta_CRCC ranking_under_BSM            ranking_under_CRCC
##       <dbl>      <dbl> <chr>                        <chr>
## 1      4.87      -11.6 Connect-and-Manage preferred Reduced exposure preferred
```

```r
# ----------------------------
# 12. Export result tables
# ----------------------------
write_csv(crcc_results, "figure5_2_crcc_results.csv")
write_csv(bsm_results, "figure5_2_bsm_results.csv")
write_csv(comparison_table, "figure5_2_comparison_table.csv")
write_csv(design_gap, "figure5_2_design_gap.csv")

# ----------------------------
# 13. Build plotting dataset
# ----------------------------
plot_data <- comparison_table %>%
  select(option_label, BSM, CRCC) %>%
  pivot_longer(
    cols = c(BSM, CRCC),
    names_to = "metric",
    values_to = "value"
  ) %>%
  mutate(
    metric = recode(
      metric,
      "BSM" = "Baseline screening metric",
      "CRCC" = "CRCC"
    ),
    option_label = factor(
      option_label,
      levels = c("Connect-and-Manage", "Associated supply / reduced exposure")
    ),
    metric = factor(
      metric,
      levels = c("Baseline screening metric", "CRCC")
    )
  )

# ----------------------------
# 14. Draw Figure 5.2
# Grouped bar chart with value labels
# ----------------------------
fig_5_2 <- ggplot(plot_data,
                  aes(x = metric, y = value, fill = option_label)) +
  geom_col(position = position_dodge(width = 0.75), width = 0.65) +
  geom_text(
```

```r
    aes(label = round(value, 1)),
    position = position_dodge(width = 0.75),
    vjust = -0.45,
    size = 4
  ) +
  scale_y_continuous(
    labels = number_format(accuracy = 0.1),
    expand = expansion(mult = c(0, 0.10))
  ) +
  labs(
    title = "Figure 5.2. Baseline screening metric versus CRCC",
    subtitle = "Decision contrast for the two stylized designs in the Virginia / PJM / DOM case",
    x = NULL,
    y = "USD per compute (lower is better)",
    fill = NULL
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold"),
    plot.subtitle = element_text(size = 11),
    legend.position = "top",
    panel.grid.minor = element_blank()
  )

print(fig_5_2)
```



```r
# ----------------------------
# 15. Save Figure 5.2
# ----------------------------
ggsave(
  filename = "Figure_5_2_BSM_vs_CRCC.png",
  plot = fig_5_2,
  width = 8.5,
  height = 5.5,
  dpi = 300
)

ggsave(
  filename = "Figure_5_2_BSM_vs_CRCC.pdf",
  plot = fig_5_2,
  width = 8.5,
  height = 5.5
)

# ----------------------------
# 16. Optional console summary
# ----------------------------
cat("\n==============================\n")
```

```
##
## ==============================
```

```r
cat("Average price used in BSM:", round(avg_price, 4), "\n")
```

```
## Average price used in BSM: 50.6929
```

```r
cat("Location-based emissions factor used in BSM:", round(g_lb, 4), "\n")
```

```
## Location-based emissions factor used in BSM: 0.2677
```

```r
cat("==============================\n\n")
```

```
## ==============================
```

```r
print(comparison_table)
```

```
## # A tibble: 2 x 13
##   option         option_label  CRCC electricity_componen~1 carbon_component_crcc
##   <chr>          <chr>        <dbl>                  <dbl>                 <dbl>
## 1 Connect-and-M~ Connect-and~  97.1                   60.9                  25.0
## 2 Associated_su~ Associated ~  85.4                   65.7                  17.5
## # i abbreviated name: 1: electricity_component_crcc
## # i 8 more variables: reliability_component_crcc <dbl>,
## #   delivered_compute_total <dbl>, potential_compute_total <dbl>,
## #   stress_hours <dbl>, BSM <dbl>, avg_price_usd_per_mwh <dbl>,
## #   location_based_emissions_tco2_per_mwh <dbl>, BSM_minus_CRCC <dbl>
```

```r
print(design_gap)
```

```
## # A tibble: 1 x 4
##   delta_BSM delta_CRCC ranking_under_BSM            ranking_under_CRCC
##       <dbl>      <dbl> <chr>                        <chr>
## 1      4.87      -11.6 Connect-and-Manage preferred Reduced exposure preferred
```